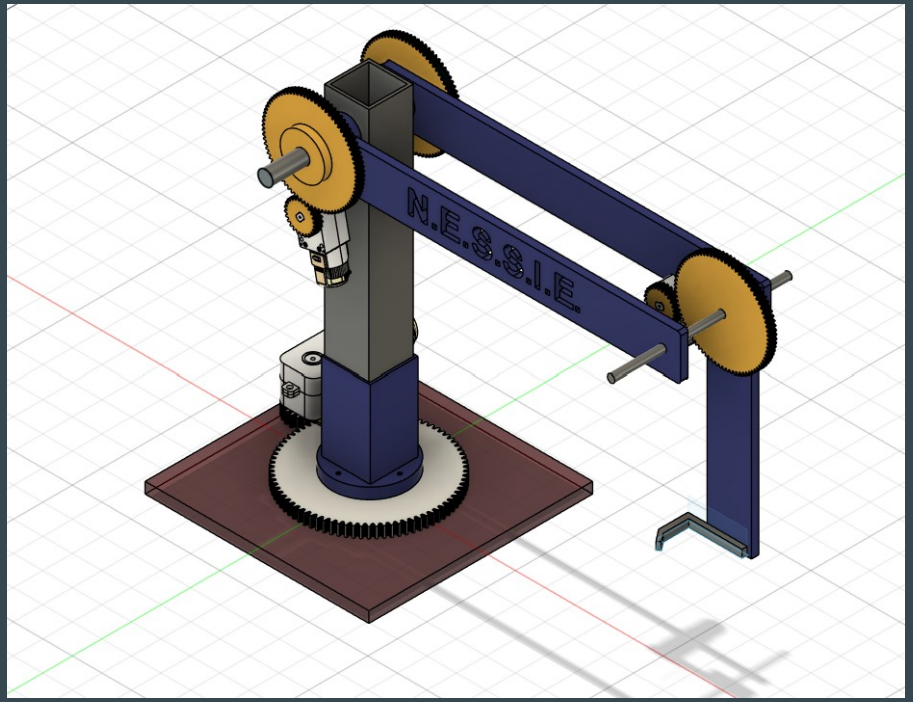
Team 27: “N.E.S.S.I.E.”





Joshua Conge, Isaac Kim, Connor Lin, Pranav Ramesh

N.E.S.S.I.E.

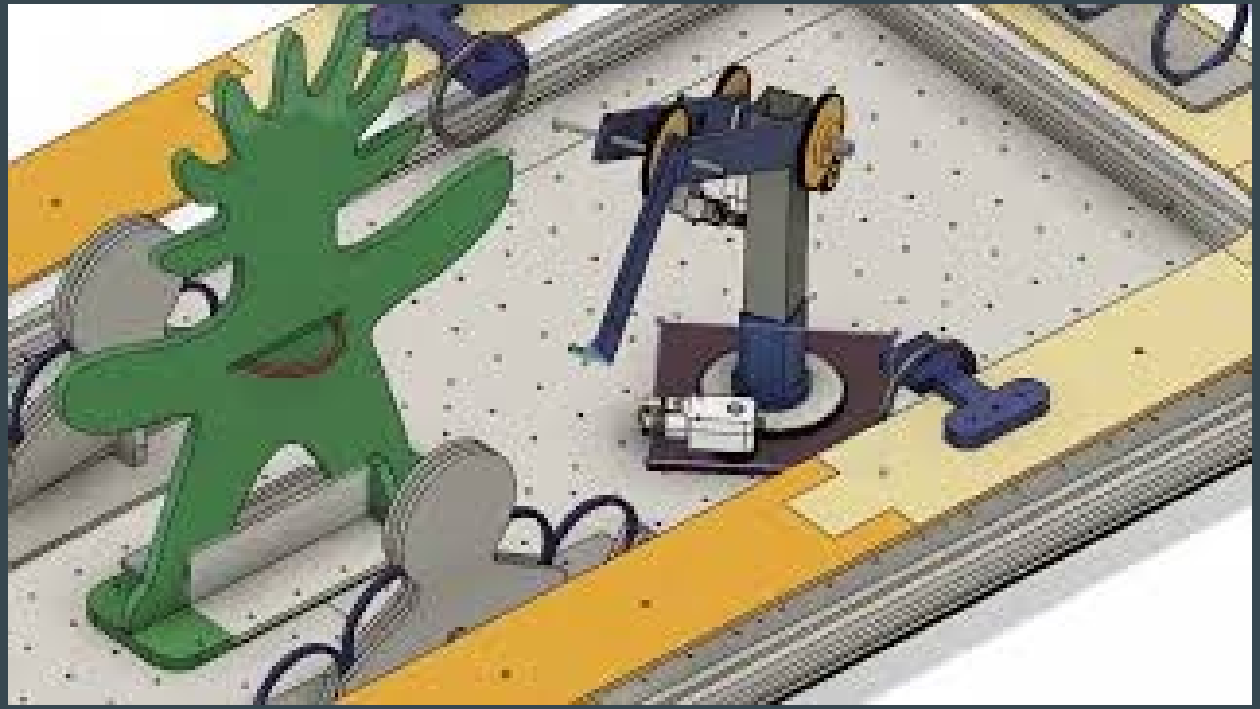


Final Design

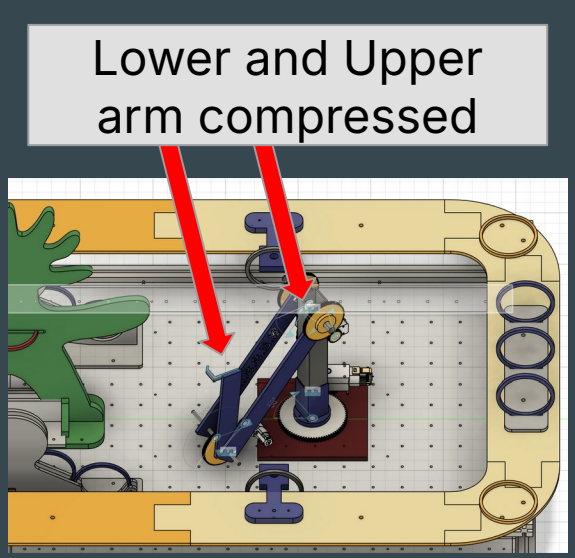
- 3 Key Powered Components
  - Rotating Base (1 geared)
  - Longer rotating Arm (2 geared)
  - Shorter rotating Arm (1 geared)

- Goals
  - Maximize scoring potential
  - Extend 18 inches from start to Sun God

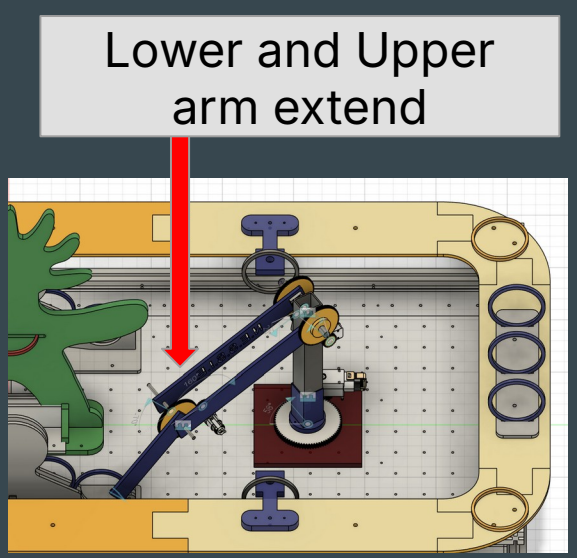
N.E.S.S.I.E. In Practice



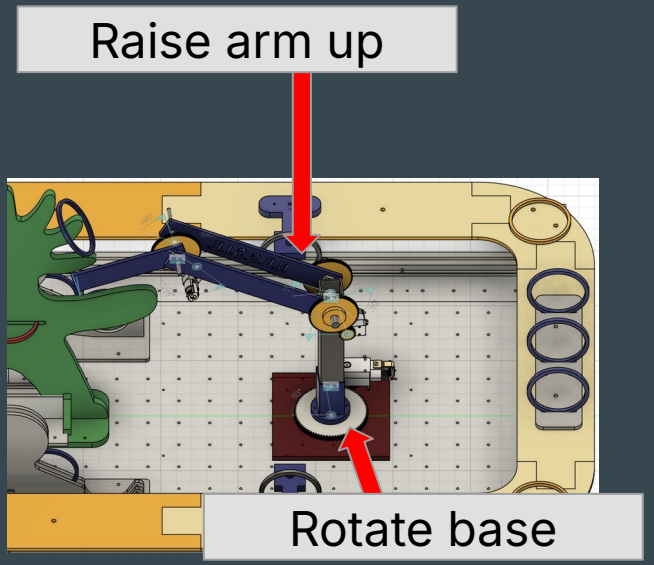
How Does N.E.S.S.I.E Work?



Step 1: Initial Position



Step 2: Reach for Rings



Step 3: Place Ring on Sungod

N.E.S.S.I.E. in Action

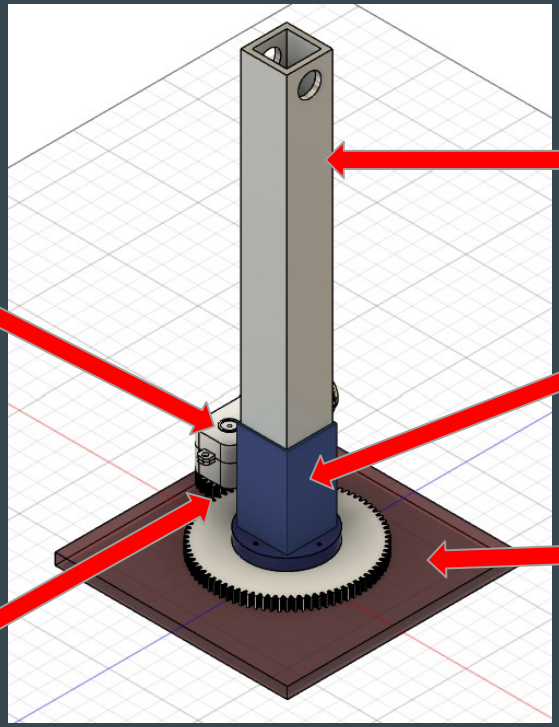
Performance:

- Max Scoring Potential: 3400 points
  - 10 rings on 3rd highest
  - 2 rings on 4th highest



- Actually Scored: 2400 points
  - 8 rings on 3rd

Base Arm



7″ Aluminum tubing

Geared Straight Motor

3D - printed holder

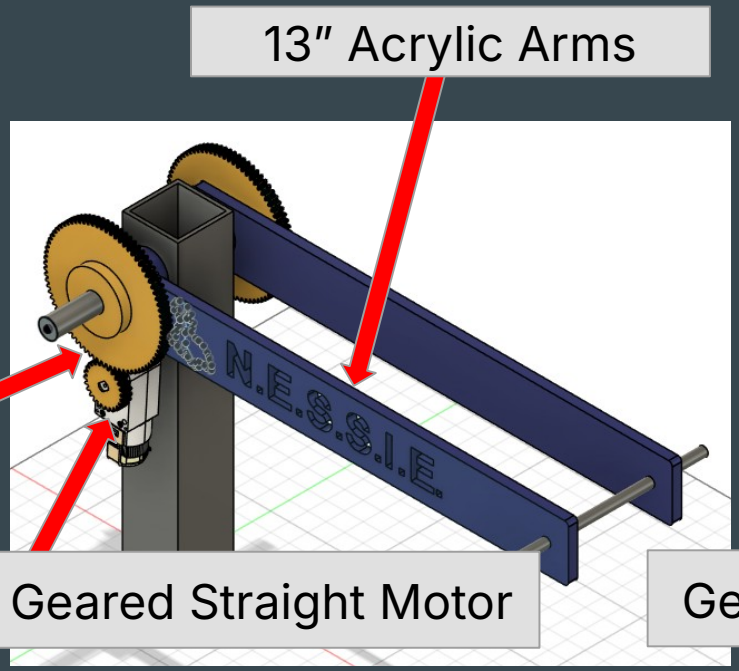
Acrylic Base Plate

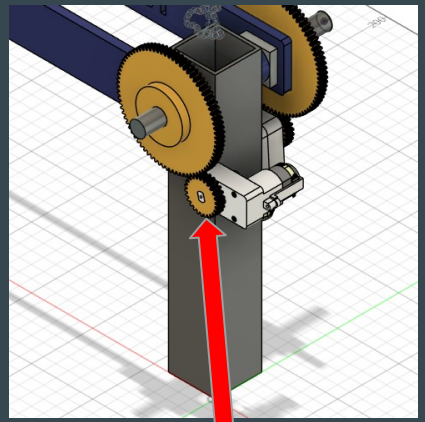
60:20 mm Gear Ratio

Functional Requirements: Rotate 650g 360°within 5s, fully extended in the horizontal position(max moment of inertia)

Lower Arm

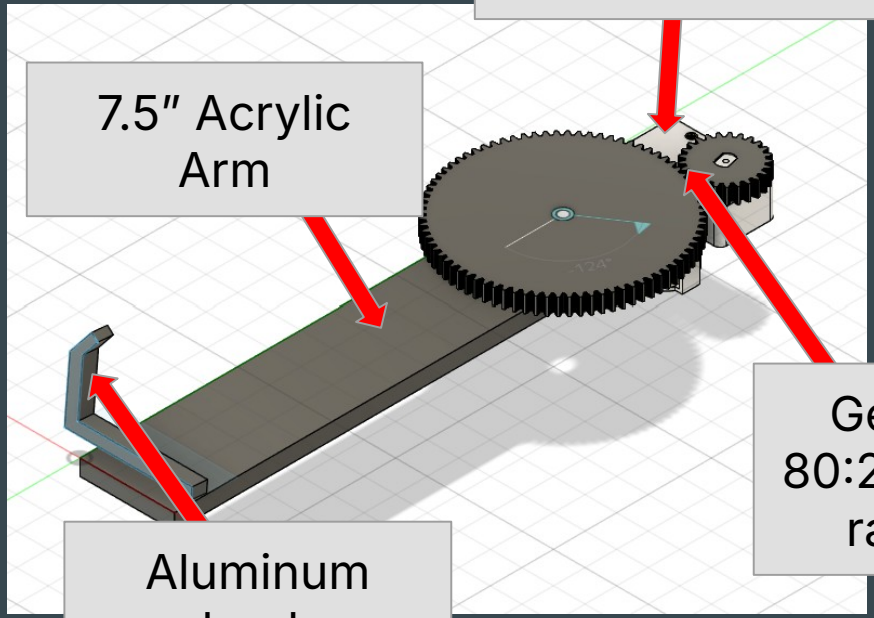
- One geared motor on each side for added to rotate arm up & down(mounted on base arm)
- Cutouts to decrease torque on the motor





Functional Requirements: Lift 200g from -45° to +90° in 7s while the upper arm is fully extended in horizontal direction

Upper Arm

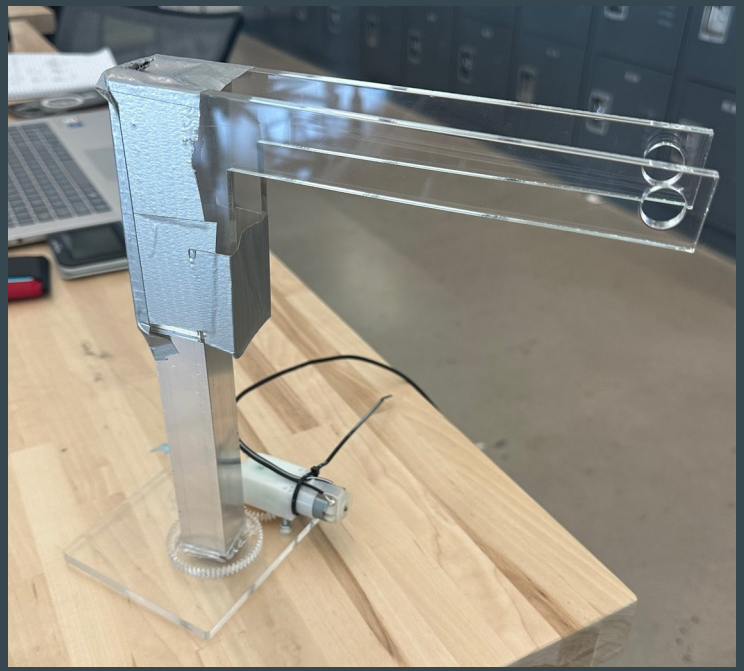


- Attaches to the end of the lower arm

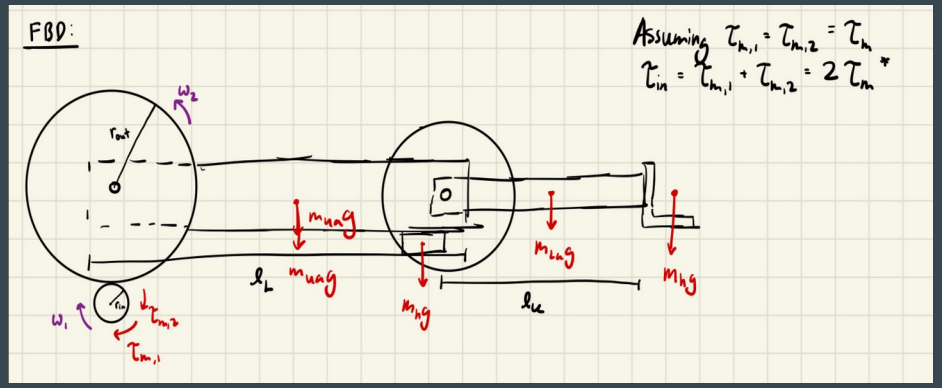
Functional Requirements: Lift from -90° to +90° in 4 seconds while carrying four rings

Risk Reduction of Base Arm

- Thought rotating base component would be riskiest
- New problem arose while manufacturing
  - Moment of weight on end of lower arm lifted the gear off the base plate

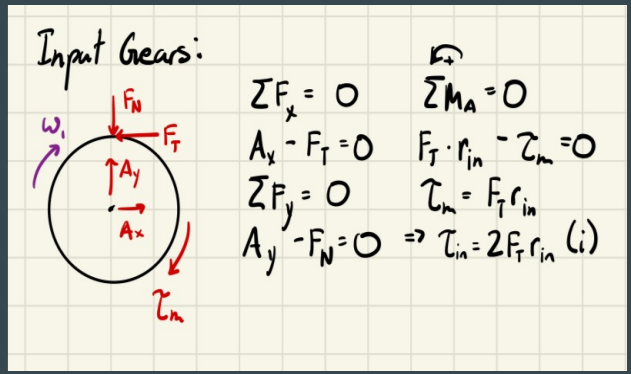


Free Body Diagrams of Lower Arm



$$\Sigma F_x = 0$$
$$F_T + B_x = 0$$
$$\Sigma F_y = 0$$
$$F_N + B_y - 2m_{La}g - m_mg - m_{ua}g - m_hg = 0$$

FBD of Arm Mechanism



FBD of Input Gears

Calculations

$$\Sigma M_A = 0$$

$$\Rightarrow F_T = \frac{\tau_{in}}{2 \cdot r_{in}} \quad (i)$$

$$\Sigma M_B = 0$$

$$\Rightarrow \tau_{in} = \frac{2 \cdot r_{in} g[2 \cdot m_{la} l_1 + m_m l_2 + m_{ua}(l_2 + l_3) + m_h(l_2 + l_3 + l_4)]}{r_{out}} \quad (ii)$$

$$FS_\tau = \frac{\tau_{in}}{\tau_{needed}} \quad (iii)$$

r_{in}= 12.5 mm

r_{out}= 40 mm

m_{la}= 102.551 g

m_{ua}= 95.603 g

m_m= 26 g

m_h= 8 g

l_1= 14.732 cm

l_2= 29.718 cm

$\Rightarrow \tau_{needed}$= .480 Nm

$\Rightarrow F_T$= 19.20 N

$\Rightarrow F.S._{torque}$= 1.13

Design: Why did we choose this?

- Simplicity
- Do not have to worry about moving around the field
- High reach; maximum height of ~20 in.
- Efficient and gets points quickly

